

# ELECTROWETTING FOR MULTI-SCALE DROPLET COALESCENCE AND TRANSPORT IN DROPWISE CONDENSATION

Joaquin Ramirez-Medina, Alidad Amirfazli\*

Department of Mechanical Engineering, York University, Toronto, M3N 3A7, Canada

## ABSTRACT

Phase-change heat transfer processes, such as condensation, are critical to water harvesting, air conditioning and power generation. While dropwise condensation offers heat transfer rates significantly higher than filmwise condensation, sustaining dropwise condensation requires precise wettability control to prevent liquid film formation. In this work, we propose and develop an active electrowetting strategy to enhance the dropwise condensation cycle. The concept is based on a dual mechanism of local condensate concentration followed by active transport. The collection mechanism creates preferred sites that promote nucleation and growth while enhancing droplet enlargement via micro-coalescence (small-scale coalescence). The transport mechanism uses sequenced electrode activation to sweep droplets away from collection zones and induce macro-coalescence (large-scale coalescence). In this way, droplets are enlarged to shedding size for removal, creating conditions for continuous surface renewal. Experiments were conducted in a closed-loop mini wind tunnel simulating a horizontally oriented condensation environment under controlled humidity and shear airflow speed. For preliminary tests, an alternating current sine signal of 120 V peak-to-peak at 250 Hz and an airflow speed of 1 m/s were used. We benchmarked our results against recent studies utilizing coplanar electrowetting configurations for dropwise condensation enhancement. Notably, our micro-macro coalescence concept accelerated droplet coarsening to shedding sizes an order of magnitude faster ( $\sim 800\ \mu\text{m}$  within  $\sim 46\ \text{s}$ ) compared to reported literature results. This enhancement was achieved under more modest operating conditions, including lower voltage, frequency, airflow velocity, and condensation potential, suggesting the potential for enhanced thermal performance under practical operating conditions.

**KEYWORDS:** Dropwise condensation, Surface renewal, Electrowetting, Shear airflow, Condensation heat transfer

## 1. INTRODUCTION

Condensation is a phase-change heat transfer process fundamental to applications such as water harvesting [1, 2], air conditioning [3], and power generation [4]. In practice, condensation can involve pure vapor or a vapor mixed with non-condensable gases (NGCs), such as humid air; this work focuses on the latter. The condensation process begins when droplet nuclei stabilize and grow on a surface, primarily by vapor diffusion [5]. As droplets enlarge, coalescence with neighboring droplets gradually supersedes diffusion as the dominant growth mechanism [6, 7]. If droplets accumulate into a continuous liquid film, the regime is termed filmwise condensation (FWC); this liquid film introduces thermal resistance that limits heat transfer. Conversely, if droplets are continuously shed before a film can form, dropwise condensation (DWC) is observed. As such, DWC is often sought because heat transfer rates can exceed those of FWC by up to an order of magnitude [8].

As an inherently cyclical process, DWC is mainly limited by how effectively and rapidly droplets can be shed from the surface; by shortening the cycle period, greater heat transfer enhancement can be achieved. To enable shedding, however, droplets must first reach a critical size at which a tangential driving force exceeds the capillary adhesion associated with contact angle hysteresis. In practice, to reach this critical size, the competing wettability requirements across the stages of the cycle must be addressed. For instance, during nucleation and

\*Corresponding Author: alidad2@yorku.ca

growth, hydrophilic regions generally promote the stability and growth of droplet nuclei, whereas during shedding, hydrophobic regions typically minimize adhesion and facilitate departure [7]. Therefore, strategies that promote DWC must not only facilitate rapid droplet coarsening to critical size, but do so without adversely affecting shedding.

Passive strategies commonly employ surface chemistry and topography to promote DWC, including superhydrophobic surfaces, liquid-infused surfaces, and hybrid wettability patterning [9-11]. However, because wettability and geometry are prescribed at fabrication, performance is often sensitive to operating conditions, e.g., the degree of subcooling and the concentration of NCGs, all of which strongly influence condensation behavior [10, 12]. More importantly, such approaches can only provide a static compromise across the stages of the DWC cycle during operation, limiting their ability to balance the competing wettability requirements of the cycle and making controlled shortening of the cycle difficult [11]. This limitation motivates active strategies that offer a route to dynamically regulate different stages of the DWC cycle to further enhance heat transfer [9, 11].

Although gravitational shedding is the most common droplet removal pathway, it is orientation-dependent and occurs only after droplets have coarsened and eventually reached shedding size. This is especially pronounced on horizontal condensing surfaces and in microgravity environments, where gravity contributes little or no tangential driving force for shedding. In such orientations, an assistive airflow can provide an aerodynamic drag force to overcome adhesion. For a first-order approximation, the drag force can be described by:

$$F_D \sim \frac{1}{2} \rho C_D A_p U_\infty^2 \quad (1)$$

where  $C_D$  is an effective drag coefficient,  $A_p$  is the projected frontal area, and  $U_\infty$  is the free-stream air velocity. As free-stream velocity increases, the boundary layer thins according to Blasius scaling ( $\delta_{99} \sim 5(vx/U_\infty)^{1/2}$ ), while droplet growth increases the projected frontal area. Accordingly, these effects can increase the drag contribution to the force balance and promote earlier shedding.

While shear airflow can supply the driving force for shedding, an active strategy remains needed to dynamically regulate and shorten the DWC cycle, thereby accelerating coarsening toward shear-assisted removal.

Electrowetting (EW) is well suited to this role as it can offer dynamic wettability control through the application of voltage. By coupling EW to a hydrophobic surface, the low-adhesion baseline favorable for the shedding stage is preserved, while voltage application promotes nucleation and coalescence, providing the dynamic wettability control across the DWC cycle stages.

Here, a capacitive system is formed between an overlapping droplet and an electrode separated by a dielectric layer. Applying an electrical potential difference across the dielectric stores electrostatic energy in the resulting electric field, with the capacitance setting the magnitude of this stored energy for a given applied voltage. Physically, this electrowetting effect manifests as a reduction in the droplet contact angle through the Young-Lippmann relation [13]:

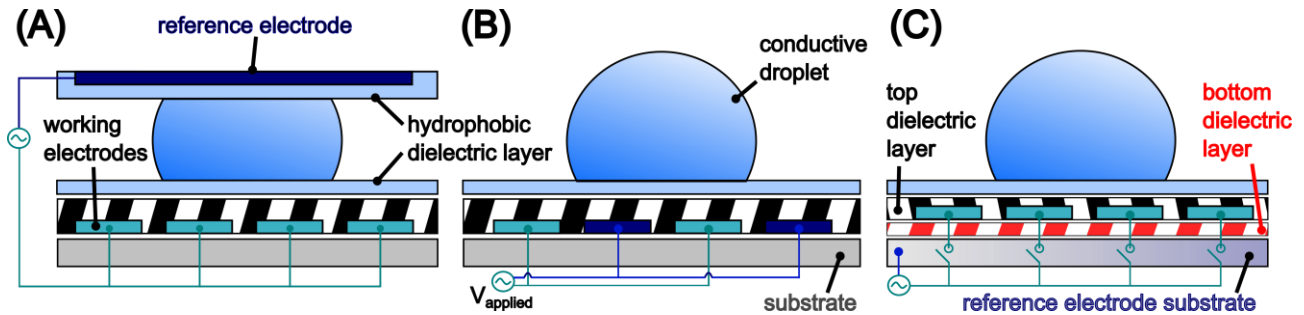
$$\cos\theta_V = \cos\theta_0 + \frac{CV^2}{2\gamma} \quad (2)$$

where  $\theta_V$  is the contact angle under applied voltage,  $\theta_0$  is the equilibrium contact angle,  $C$  is the capacitance per unit area,  $\gamma$  is the liquid-vapor interfacial tension, and  $V$  is the applied voltage.

However, Eq. 2 describes the local wettability change associated with the applied voltage and capacitance. In practice, whether this change manifests only as local static spreading or instead induces droplet motion depends on the spatial distribution of the electric field. For instance, if the field remains uniform, only static spreading

is observed. When the electric field is spatially non-uniform, an electrostatic energy gradient is created, driving droplet motion via electrocapillary forces toward surface free energy minima.

As such, the EW configuration – that is, the electrode geometry and arrangement, dielectric thickness and properties – governs the capacitance magnitude and spatial distribution of the resulting electric field, which determines the extent of wettability change and the resulting droplet motion.



**Fig. 1** Displays common EW configurations, including (A) closed-surface parallel-plate and (B) open-surface, coplanar layouts. (C) Depicts an open-surface parallel-plate configuration, with electrodes interposed between two dielectric layers.

One common EW configuration is the closed-surface, parallel-plate layout (Fig. 1A). In this configuration, droplets are confined between working and reference electrodes, producing vertical electric fields [14-20]. Since droplets capacitively couple across two dielectric interfaces, greater electrostatic energy can be stored. When combined with electrode geometries that produce spatially non-uniform fields, strong energy gradients can be generated that drive droplet movement. For this reason, various electrode geometries, e.g., chevron and heart shapes, have been previously used in digital microfluidics for mixing, splitting, and stepwise directional transport of single droplets [14, 16, 18, 20].

Another common configuration includes coplanar layouts (Fig. 1B) where electrodes are arranged on a single plane with alternating polarity, exposing the surface and the droplets to the surrounding vapor. This open-surface setup enables a broader range of studies, including self-cleaning surfaces [21] and condensation. In condensation studies, coplanar configurations have often been studied in vertical orientations, where EW primarily targets coalescence enhancement, rather than nucleation. Two electrode geometries commonly employed in condensation settings are straight interdigitated electrodes [22-25] and triangular interdigitated electrodes [25, 26]. Unlike parallel-plate configurations, straight interdigitated electrodes produce energy gradients across electrode gaps, creating surface free energy minima at gap centers, towards which growing droplets are driven via coalescence. Triangular interdigitated electrodes use geometric asymmetry to produce energy gradients along electrode edges. Combined with coalescence, these gradients drive growing droplets towards gap apices, where surface free energy is minimized. In both geometries, droplets gradually accumulate at energy minima until reaching critical shedding size for gravity-assisted shedding.

A key limitation of coplanar configurations arises from their reliance on lateral electric fields. Since these fields become weaker with increasing electrode pitch and dielectric thickness, the electrostatic energy available for actuation is limited [13]. Consequently, as per the Young-Lippmann relation, greater voltage is required to achieve a wettability change comparable to that of parallel-plate layouts. Additionally, because lateral field asymmetry must be imposed through the periodic electrode polarity, the geometry is constrained to relatively simple, rigid patterns. This makes it challenging to implement more complex geometries that could further enhance the DWC cycle.

Among the coplanar studies, the work conducted in [27] is most relevant to the present study as it employed a similar closed-loop wind tunnel to analyze thermal performance. An AC voltage of 424 V<sub>pp</sub> at 1 and 10 kHz

was applied to straight interdigitated electrodes on a horizontal, subcooled surface, held at 0 °C. A low airflow velocity of 1 m/s and relative humidity (RH) of 60 % were used to visualize early-stage drop morphology, while higher velocities from 5 to 15 m/s were used to study shear-driven shedding and the associated thermal performance. Increasing airflow velocity, voltage, and frequency enhanced droplet shedding and increased the overall heat transfer coefficient (HTC). Specifically, the 10 kHz case improved the HTC by 36 % and 38.5 % at airflow velocities of 10 and 15 m/s, respectively, compared to the control. Nevertheless, because growth still relied on coalescence-driven enlargement, droplet residence times remained prolonged.

Alternatively, an open-surface parallel-plate configuration (Fig. 1C) has been previously explored, where EW can target the nucleation stage in addition to growth. In this layout, the working electrode is interposed between two distinct dielectric layers, and the substrate is the reference electrode [28-32]. Unlike closed-surface, parallel-plate layouts, droplets couple through a single dielectric interface; and unlike coplanar configurations, the electric fields are directed vertically into the substrate rather than bridging laterally across electrode gaps. Here, grid-wire electrodes concentrate these fields along grid edges, creating localized energy minima that can lower the nucleation energy barrier and promote nucleation and growth [26]. As droplets enlarge, they migrate along energy gradients toward grid cell interiors, thereby clearing edges for re-nucleation. However, cell interiors act as gradual accumulation regions where droplets reside until they are large enough to capacitively couple with neighboring grid edges and move toward grid centers. Once there, droplets continue to grow until eventually reaching critical shedding sizes for gravity-assisted shedding [29-32].

Despite these distinctions, all of the above configurations share a common bottleneck: coalescence-driven coarsening remains gradual, which prolongs droplet residence on the condensation surface. The open-surface parallel-plate layout offers a unique advantage because its resulting field can promote nucleation and early-stage growth, capabilities that coplanar layouts lack. However, once droplets migrate to grid-cell interiors, growth reverts to gradual coalescence without a mechanism to accelerate coarsening toward shedding size.

To overcome this bottleneck, a more effective electrowetting strategy should preserve the nucleation advantages of the open-surface parallel-plate layout while actively accelerating the downstream stages of the DWC cycle. An ideal DWC cycle would: (i) enhance nucleation and growth by triggering small-scale coalescence, (ii) transport droplets away to continuously renew nucleation and growth regions, (iii) facilitate a second intermediate-scale coalescence during transport, and (iv) shed droplets at reduced timescales. Together, these functions would shorten the cycle period and enhance the HTC [11, 34].

In this work, we employ an open-surface, parallel-plate configuration with a novel electrode geometry and arrangement designed to address the above points, thereby shortening the DWC cycle. The design integrates a dual collection and transport mechanism. The collection mechanism uses specific electrodes to enhance nucleation and small-scale, “micro-coalescence”, i.e., coalescence at the nanometer to several micrometers range. The transport mechanism is adapted from digital microfluidics and uses stepwise activation to clear the collection areas; this mechanism targets “macro-coalescence” at the several tens to hundreds of micrometers range, up to the shedding stage where droplets are typically millimetric.

To simulate a practical condensation environment, representative of horizontally oriented applications, we employ a closed-loop mini wind tunnel. The wind tunnel enables control of airflow speed, humidity, and temperature for setting the degree of condensation and the shear airflow conditions. Although silicon substrates were used here to facilitate microfabrication, other substrates can be employed in practical applications. We then comment on and compare the proposed design to previous electrowetting-assisted DWC studies.

## 2. METHODS

**2.1 Electrowetting System.** An open-surface, parallel-plate configuration was fabricated to generate strong vertical electrical fields to enhance the DWC cycle. A boron-doped, 500  $\mu\text{m}$  thick conductive silicon substrate ( $\rho \approx 0.001\text{--}0.005 \text{ }\Omega\cdot\text{cm}$ ) served as the reference electrode. A 100 nm silicon nitride bottom dielectric layer was deposited via low-pressure chemical vapor deposition (LPCVD). Positive photoresist (S1813) was spin-

coated to a thickness of  $\sim 1.8 \mu\text{m}$  using a two-step cycle: 500 rpm (100 rpm/s ramp) for 5 s, followed by 2000 rpm (1000 rpm/s ramp) for 30 s. Patterns were transferred via UV exposure (OAI Model 800E, 150 W) through a mask for  $\sim 14$  s, followed by development in MF-319 for 45 s. A 5 nm Ti adhesion layer and a 50 nm Au layer were deposited via e-beam evaporation (Angstrom Engineering Nexdep). Lift-off was performed in PG Remover (80 °C bath for 45 min), followed by two 10 s sonication pulses. Electrode pad areas were covered with masking tape to avoid any subsequent deposition in those regions. A 150 nm silicon nitride top dielectric layer was deposited via PECVD (Trion Technology Minilock Phantom Duo). The silicon wafer was diced into 15 mm x 25 mm coupons to fit within the wind tunnel test section of 75 mm x 25 mm. Finally, a hydrophobic top layer (Cytonix Fluoropel PFC1601VF-S) was spin-coated using the same recipe as above for S1813 and baked at 180 °C for 30 min to drive down surface energy. The masking tape was peeled, exposing the electrode pad contact points for electrical connection to the electrowetting chip.

**2.2 Experimental Setup.** Condensation experiments were conducted in a closed-loop mini wind tunnel allowing independent control of relative humidity (RH), air velocity, and substrate temperature. The test section (150 mm long, 50 mm  $\times$  50 mm cross-section) housed the EW substrate mounted flush with the floor of the test section. For these preliminary tests, the operating conditions were held at 60 % RH, 1 m/s air velocity, and a substrate temperature of 5 °C. The airflow speed of 1 m/s circulated fresh humid air over the surface, thinning the non-condensable gas (NCG) boundary layer.

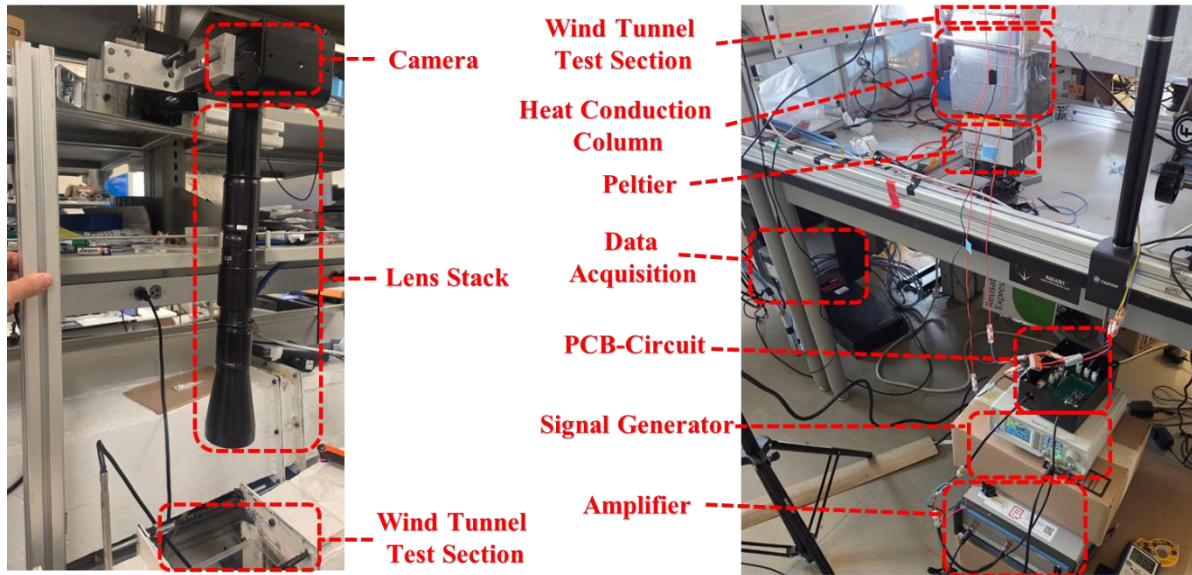
The presence of such an NCG boundary layer is known to significantly reduce condensation rates [35]. In the absence of a strong shear airflow, significant aerodynamic forces are not induced on the droplets; therefore capillary forces remained dominant. Using lower airflow speeds allowed for observing droplet growth over time without any interference from shedding. A RH of 60 % was sufficient to promote condensation, yet low enough to prevent fog formation on the inner wind tunnel walls, which would obstruct visualization.

A PID humidity controller (HD100, Auber) regulated the moisture content in the loop and provided RH readings. Airflow was generated by an axial fan (4114 NH6, ebm-papst) powered by a bench supply and controlled via a pulse width modulation driver. The mean air velocity in the test section was measured using a hot-film anemometer (EE75, E+E Elektronik GmbH). Substrate temperature was controlled using a thermoelectric Peltier module (CP-031, TE Technology) mounted beneath the meter bar and actively cooled. Clear depictions of the complete wind tunnel setup and configuration can be found in [27, 33].

Figure 2 displays the main instrumentation setup, including an overhead high-speed camera (Chronos 2.1-HD) and Navitar lens stack to capture droplet morphology (field of view  $\sim 3.2 \times 1.6 \text{ mm}^2$ ). To enable on-demand electrode activation, electrode columns were connected to solid-state relays (SSRs) on a PCB-circuit, as shown in the figure. SSRs were driven using a function generator (Koolertron DDS) and a high-voltage amplifier (Pendulum F20A) producing sinusoidal AC voltages. To limit charge buildup and increase the longevity of the electrowetting devices, DC signals were not used.

Electrical connection to on-chip electrode pads was made using 30 AWG wires bonded with conductive epoxy (MG Chemicals 8331S). The joints and exposed wire segments were covered with epoxy resin (MG Chemicals 832XFC) to provide electrical and moisture insulation, as well as strain relief. For all experiments discussed in this paper, the electrodes were driven with a sinusoidal AC waveform of 120 V<sub>pp</sub> at 250 Hz. The waveform signal was routed through the SSRs channels to corresponding electrode columns and controlled via a developed graphical user interface.

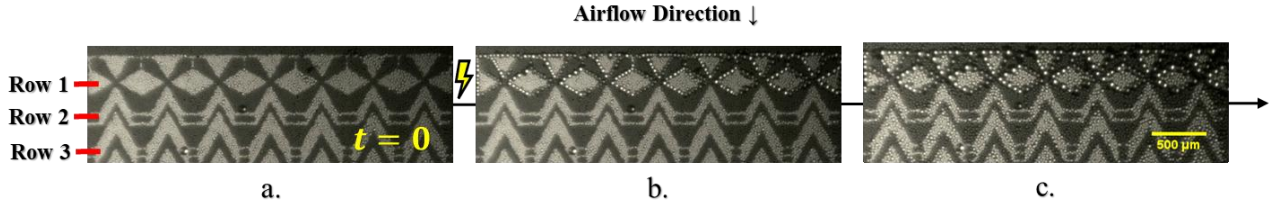
The electrode layout was designed based on a first-order electrowetting actuation model to predict the net lateral actuation force as a function of droplet position and applied voltage for a representative droplet size.



**Fig. 2** Instrumentation for the experimental setup; the wind tunnel is not shown.

### 3. RESULTS AND DISCUSSION

**3.1 Droplet Morphology.** We analyze droplet growth dynamics over time without shedding, as this can provide an indication of the effectiveness of the dual mechanism, namely, the collection and transport mechanism. Below, we present two separate time sequences, one for the collection mechanism and one for active transport, illustrating how the dual mechanism can shorten the DWC cycle period.



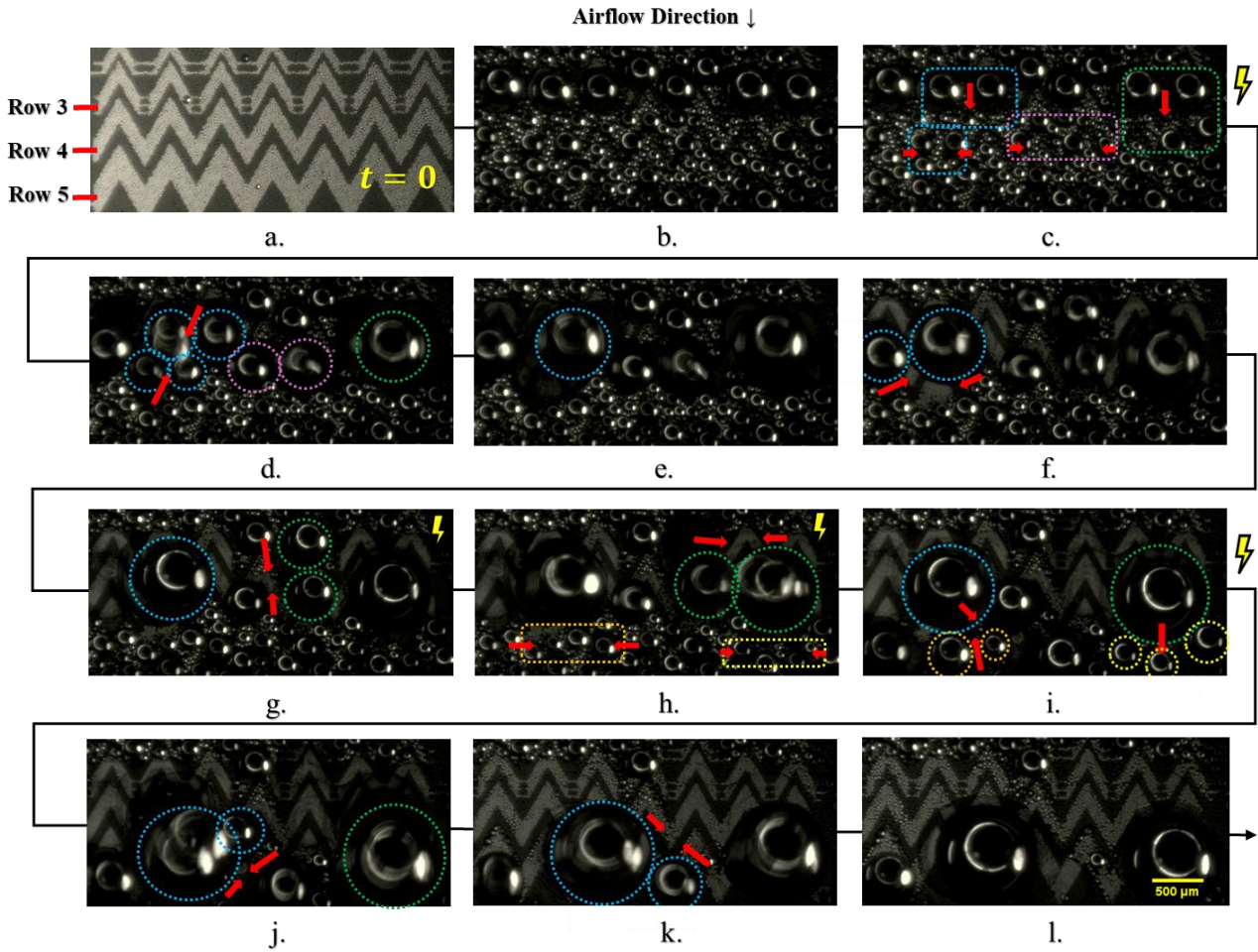
**Fig. 3** Time sequence of the collection mechanism after activation of electrode row 1. a) Condensing surface before activation, b) immediately after activation, and c)  $\sim 5$  s after activation.

A time sequence for the collection mechanism is shown in Fig. 3, highlighting early-stage droplet enlargement through micro-coalescence. Figure 3a shows the condensation surface prior to activation of electrode row 1. Figure 3b shows the surface after activation of row 1, where droplets have grown to diameters of  $\sim 20$   $\mu\text{m}$ . Approximately 5 s after activation of row 1 (Fig. 3c), the droplets have doubled in diameter to  $\sim 40$   $\mu\text{m}$ . Here, enlargement is mainly attributed to repeated micro-coalescence events occurring along the activated electrode edges due to the alternating nature of the voltage used. While droplets enlarge on active electrode areas (i.e., along the edges of electrode row 1), droplets in the non-active regions (i.e., electrode rows 2 and 3) remain significantly smaller ( $< 20$   $\mu\text{m}$ ), providing evidence for the proposed mechanism.

In the literature discussed above, similar small-scale droplet growth occurred in a more random manner on the condensation surface and was not actively promoted by coalescence. Conversely, in our configuration, micro-coalescence provides a rapid enlargement boost, coarsening small-scale droplets in targeted regions within a few seconds, after which they can transition to the transport mechanism. In this sense, the collection mechanism acts as a controllable pre-conditioning stage that quickly converts numerous nucleation-scale droplets into fewer, larger droplets that are better suited for efficient downstream transport and eventual shedding.



Figure 4 illustrates the transport mechanism and how macro-coalescence further enlarges droplets and moves them away from collection zones, thereby shortening the DWC cycle. This sequence shows how the transport mechanism addresses the limitations of previous coplanar configurations, in which coalescence was gradual. Color-coded encirclements are used to track droplet groups that undergo macro-coalescence during transport from electrode row 3 to 5.



**Fig. 4** Time sequence of the transport mechanism for electrode rows 3 – 5. Frames (b-l) span a total duration of  $\sim 46$  s, over which droplets grow to diameters of  $\sim 800$   $\mu\text{m}$ .

Initially, electrode row 3 is deactivated (Fig. 4b–c), and droplets on this row have diameters between 250 – 350  $\mu\text{m}$ . Upon activation of row 4 (Fig. 4d), droplets rapidly shift to maximize capacitive coupling, triggering an immediate macro-coalescence event. Figure 4e displays the frame immediately after this event, with electrode row 4 still activated. Figure 4f shows the frame just before another macro-coalescence event that forms the blue-encircled droplet in Fig. 4g. Between Fig. 4g and Fig. 4i, the signal on row 4 is pulsed twice. The first pulse (Fig. 4g) induces macro-coalescence of the smaller droplets as seen in green-encircled areas. The second pulse (Fig. 4h) induces another macro-coalescence of droplets, as seen in the orange-, yellow-, and green-encircled droplet groups, producing larger droplets. Finally, electrode row 5 is activated (Fig. 4i–j), and the droplets shift downwards towards the lower surface free energy areas, where they undergo final macro-coalescence events that further enlarge the droplets, see blue- and green-encircled drops. The signal on row 5 is then maintained (Fig. 4j–k) before being switched off (Fig. 4l). Notably, droplets on row 5 reach diameters of  $\sim 800$   $\mu\text{m}$  within a total duration of  $\sim 46$  s, including activation, deactivation, and pulse times.

Overall, the transport sequence demonstrates that stepwise activation can deliberately orchestrate macro-coalescence and motion, coarsening multiple intermediate-sized droplets into fewer, shedding-scale droplets while simultaneously clearing upstream regions for renewed nucleation and growth.

To contextualize these observations, we next compare the operating conditions across studies in terms of condensation potential, i.e., the degree to which condensation is thermodynamically driven. In the presence of NCGs, condensation is controlled by a gas–vapor diffusion boundary layer that depends on the dewpoint subcooling difference ( $\Delta T_{\text{dew}}$ ) and on the water vapor partial pressure difference ( $\Delta p_v$ ) between the humid air and the substrate surface [35]. Accordingly, for each condensation condition across the previous and current work, we discuss how the operating parameters influence the degree of condensation. This comparison is important because nucleation-to-shedding cannot be interpreted solely in terms of electrode geometry and arrangement; it must also be understood relative to how strongly each setup drives condensation.

As summarized in Table 1, the previous coplanar-configuration studies operated at higher condensation potential than the present work, indicated by larger  $\Delta T_{\text{dew}}$  and/or  $\Delta p_v$ . Studies [24] and [26] worked near or at saturated air conditions providing a strong degree of condensation. Study [27] matched the current air conditions ( $T_{\text{air}}$  and RH), but used a colder surface, yielding a higher condensation potential than in the present study. Thus, the present work operates with the lowest condensation driving force among the studies compared.

**Table 1** Comparison of environmental conditions for condensation in previous studies and current work.

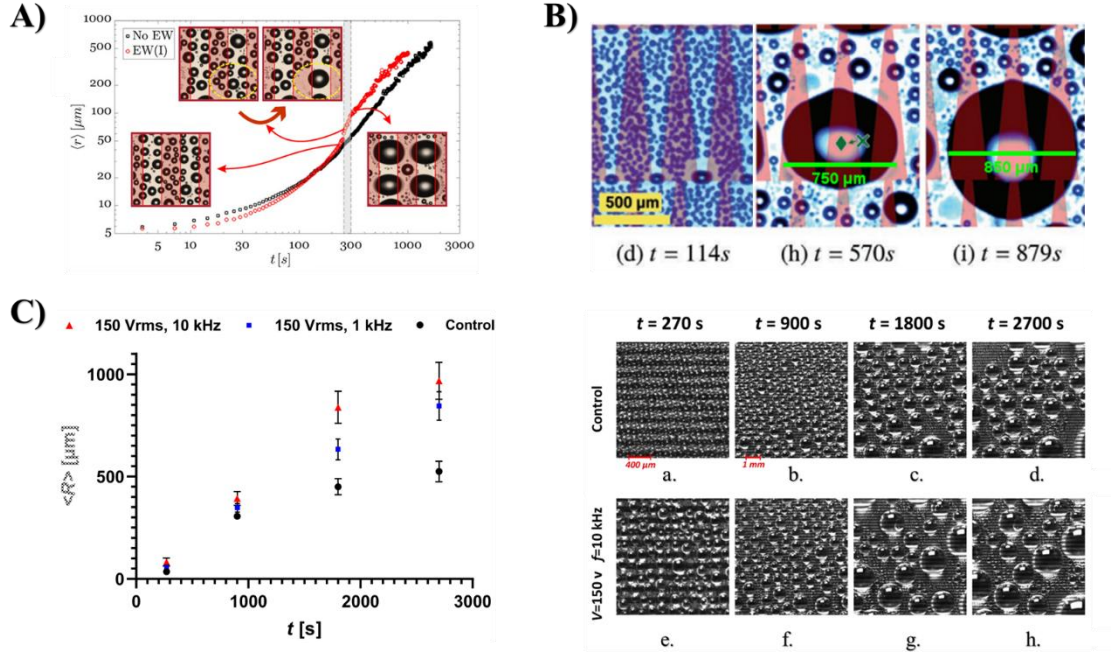
| Study                | $T_{\text{air}}$ (°C) / RH (%) | $T_{\text{surf}}$ (°C) | $\Delta T_{\text{dew}}$ (°C) | $\Delta p_v$ (kPa) |
|----------------------|--------------------------------|------------------------|------------------------------|--------------------|
| Baratian et al. [24] | 36 / 100                       | 25                     | ~ 11                         | ~ 2.8              |
| Hoek et al. [26]     | 42 / 99                        | 11.5                   | ~ 30                         | ~ 6.9              |
| Bonab et al. [27]    | 25 / 60                        | 0                      | ~ 17                         | ~ 1.3              |
| <b>Current study</b> | <b>25 / 60</b>                 | <b>5</b>               | <b>~ 12</b>                  | <b>~ 1.0</b>       |

Next, we contrast the results of the present transport mechanism with the results of the previous coplanar configurations. To compare these studies, we analyze how quickly droplets in other works reach a diameter of  $\sim 800 \mu\text{m}$ , which is also summarized in Table 2 for clarity. Figure 5 displays a compilation of droplet growth over time from the previous studies for straight interdigitated electrodes (Fig. 5a and Fig. 5c) and triangular interdigitated electrodes (Fig. 5b).

For straight interdigitated electrodes, droplets reach a comparable size after  $\sim 850$  s in (Fig. 5a) and after  $\sim 900$  s in (Fig. 5c). For triangular interdigitated electrodes, a faster growth is evident with droplets reaching a comparable size after  $\sim 570$  s. Although the triangular interdigitated electrodes (Fig. 5b) demonstrate faster growth than the straight interdigitated electrodes (Fig. 5a and Fig. 5c), all prior configurations still required growth times roughly an order of magnitude longer than the current study to reach  $\sim 800 \mu\text{m}$ . The above illustrates the coalescence enhancements that the transport mechanism achieves, despite operating under the lowest degree of condensation.

We also compare the current study with [27], which uses a similar wind tunnel setup, but we focus on the operating conditions related to the wind tunnel and the applied signal. In this previous work, airflow velocities from 5 to 15 m/s were used for droplet shedding where incipient motion began at a critical shedding radius of  $\sim 400 \mu\text{m}$ . The higher airflow velocities used in [27] not only increased the drag force available for shedding, but also further thinned the NCG boundary layer. By increasing airflow velocity, vapor transport to the surface is enhanced and the degree of condensation increases. In particular, we consider the fastest shedding case at 15 m/s airflow velocity, where incipient motion occurred after  $\sim 56$  s using an applied voltage of  $424 V_{\text{pp}}$  at 10 kHz. Conversely, the transport mechanism enables droplets to reach the same critical shedding radius of  $\sim 400 \mu\text{m}$ , yet after  $\sim 46$  s while subjected to a much lower airflow velocity (1 m/s) and a substantially weaker EW signal ( $120 V_{\text{pp}}$  at 250 Hz).





**Fig. 5** Growth dynamics of droplets in previous works. Insets show droplet sizes as a function of time for A) Baratian et al., [24] B) Hoek et al. [26] and C) Bonab et al. [27]. Images are used with permission.

Together, these results show that the collection and transport mechanisms can reduce the time to critical shedding size under much more modest operating conditions. The resulting shorter DWC cycle time reduces the residence time of droplets on the surface, which is a key factor governing the effective HTC in condensation systems. Although we do not directly measure the HTC in the present work, the substantial reduction in growth time to shedding sizes strongly suggests an increase in effective HTC.

**Table 2** Comparison of shortest droplet growth times to  $\sim 800 \mu\text{m}$  reported in previous studies and the current work, including airflow speed ( $U$ ) and actuation conditions ( $V_{pp}$  and  $f$ ).

| Study                | Time to $\sim 800 \mu\text{m}$ (s) | $U$ (m/s) | $V_{pp}$ (V) | $f$ (Hz)   |
|----------------------|------------------------------------|-----------|--------------|------------|
| Baratian et al. [24] | $\sim 900$                         | —         | $\sim 424$   | 1000       |
| Hoek et al. [26]     | $\sim 570$                         | —         | $\sim 424$   | 1000       |
| Bonab et al. [27]    | $\sim 900$                         | 1         | $\sim 424$   | 10000      |
| <b>Current study</b> | <b><math>\sim 46</math></b>        | <b>1</b>  | <b>120</b>   | <b>250</b> |

## 4. CONCLUSIONS

In this work, we demonstrate a micro–macro coalescence and transport strategy that can shorten the DWC cycle by an order of magnitude compared with previous works. For this purpose, we designed an electrode geometry and arrangement that integrates a dual collection and transport mechanism where early-stage droplets are enlarged through micro-coalescence at collection sites. During transport, macro-coalescence sweeps droplets away from collection sites while coarsening them to shedding sizes. Notably, the strategy was demonstrated on a horizontal condensing surface, where gravity provides no tangential driving force for shedding; this indicates applicability to vertical and other orientations where gravity can further assist removal.

The current study operated at milder operating conditions than previous works, including a lower condensation potential ( $\Delta T_{\text{dew}} \approx 11.7 \text{ }^\circ\text{C}$  and  $\Delta p_v \approx 1.03 \text{ kPa}$ ), a lower applied AC signal ( $120 V_{pp}$  at  $250 \text{ Hz}$ ), as well as

lower airflow velocity (1 m/s). Despite these conditions, droplets reached comparable shedding sizes of ~800  $\mu\text{m}$  in diameter within ~46 s. The substantial reduction in cycle time suggests an enhancement in HTC.

## ACKNOWLEDGMENTS

The authors acknowledge funding from the Natural Sciences and Engineering Research Council of Canada (NSERC).

## REFERENCES

- [1] Lee, A., et al., “Water harvest via dewing”, *Langmuir*, 28(27), pp. 10183-10191, (2012).
- [2] Hou, Y., et al., “Recurrent filmwise and dropwise condensation on a beetle mimetic surface”, *ACS Nano*, 9(1), pp. 71-81, (2015).
- [3] Cavallini, A., et al., “Condensation inside and outside smooth and enhanced tubes—a review of recent research”, *Int. J. Refrig.*, 26(4), pp. 373-392, (2003).
- [4] Chuang, C.C. and Sue, D.C., “Performance effects of combined cycle power plant with variable condenser pressure and loading”, *Energy*, 30(10), pp. 1793-1801, (2005).
- [5] Rose, J.W., “On the mechanism of dropwise condensation”, *Int. J. Heat Mass Transf.*, 10(6), pp. 755-762, (1967).
- [6] Meakin, P., “Steady state behavior in a model for droplet growth, sliding and coalescence: the final stage of dropwise condensation”, *Physica A*, 183(4), pp. 422-438, (1992).
- [7] Cho, H.J., et al., “Nanoengineered materials for liquid–vapour phase-change heat transfer”, *Nat. Rev. Mater.*, 2(2), pp. 1-17, (2016).
- [8] Reddy, M.R.N., Yohan, M. and Reddy, K.H., “Heat transfer co-efficient through dropwise condensation and filmwise condensation apparatus”, *Int. J. Sci. Res. Publ.*, 2(12), pp. 1-4, (2012).
- [9] Sun, J. and Weisensee, P.B., “Tailoring wettability to push the limits of condensation”, *Curr. Opin. Colloid Interface Sci.*, 67, p. 101739, (2023).
- [10] Goswami, A., Pillai, S.C. and McGranaghan, G., “Surface modifications to enhance dropwise condensation”, *Surf. Interfaces*, 25, p. 101143, (2021).
- [11] Hughes, M.T. and Garimella, S., “A review of active enhancement methods for boiling and condensation”, *Int. J. Heat Mass Transf.*, 218, p. 124752, (2024).
- [12] Huang, J., Zhang, J. and Wang, L., “Review of vapor condensation heat and mass transfer in the presence of non-condensable gas”, *Appl. Therm. Eng.*, 89, pp. 469-484, (2015).
- [13] Mugele, F. and Heikenfeld, J., *Electrowetting: Fundamental Principles and Practical Applications*, Hoboken: John Wiley & Sons, (2019).
- [14] Rajabi, N. and Dolatabadi, A., “A novel electrode shape for electrowetting-based microfluidics”, *Colloids Surf. A*, 365(1-3), pp. 230-236, (2010).
- [15] Dong, C., et al., “On the droplet velocity and electrode lifetime of digital microfluidics: voltage actuation techniques and comparison”, *Microfluid. Nanofluid.*, 18(4), pp. 673-683, (2015).
- [16] Wang, W., et al., “An asymmetric electrode for directional droplet motion on digital microfluidic platforms”, *Sens. Actuators B Chem.*, 324, p. 128763, (2020).
- [17] Lin, Y.Y., Welch, E.R.F. and Fair, R.B., “Low voltage picoliter droplet manipulation utilizing electrowetting-on-dielectric platforms”, *Sens. Actuators B Chem.*, 173, pp. 338-345, (2012).
- [18] Xu, X., et al., “Electrowetting on dielectric device with crescent electrodes for reliable and low-voltage droplet manipulation”, *Biomicrofluidics*, 8(6), p. 064114, (2014).
- [19] Shen, H.H., et al., “EWOD microfluidic systems for biomedical applications”, *Microfluid. Nanofluid.*, 16(5), pp. 965-987, (2014).
- [20] Nahar, M.M., et al., “Droplet velocity in an electrowetting on dielectric digital microfluidic device”, *Micromachines*, 7(4), p. 71, (2016).
- [21] Sun, D., Gomez, G. and Böhringer, K.F., “Droplet manipulation using AC Ewod-actuated anisotropic ratchet conveyor”, *Proc. of 20th Int. Conf. on Solid-State Sensors, Actuators and Microsystems (TRANSDUCERS & EUROSENSORS XXXIII)*, pp. 120-123, (2019).
- [22] Wikramanayake, E.D., Perry, J. and Bahadur, V., “AC electrowetting promoted droplet shedding on hydrophobic surfaces”, *Appl. Phys. Lett.*, 116(19), p. 193701, (2020).
- [23] Wikramanayake, E.D. and Bahadur, V., “Electrowetting-based enhancement of droplet growth dynamics and heat transfer during humid air condensation”, *Int. J. Heat Mass Transf.*, 140, pp. 260-268, (2019).
- [24] Baratian, D., et al., “Breath figures under electrowetting: electrically controlled evolution of drop condensation patterns”, *Phys. Rev. Lett.*, 120(21), p. 214502, (2018).
- [25] Dey, R., et al., “Controlling shedding characteristics of condensate drops using electrowetting”, *Appl. Phys. Lett.*, 113(24), p. 243701, (2018).
- [26] Hoek, H., Dey, R. and Mugele, F., “Electrowetting - controlled dropwise condensation with patterned electrodes: physical principles, modeling, and application perspectives”, *Adv. Mater. Interfaces*, 8(2), p. 2001317, (2021).

- [27] Bonab, M.S., et al., “Electrowetting assisted shedding of droplets by a shearing airflow in dropwise condensation”, *Int. J. Heat Mass Transf.*, 220, p. 124925, (2024).
- [28] Cooney, C.G., et al., “Electrowetting droplet microfluidics on a single planar surface”, *Microfluid. Nanofluid.*, 2(5), pp. 435-446, (2006).
- [29] Pham, R., et al., “Condensations on electrowetting surfaces: An electrostatic liquid conveyor from dropwise-filmwise competition”, *Int. J. Heat Mass Transf.*, 213, p. 124280, (2023).
- [30] Yan, R. and Chen, C.L., “Condensation droplet distribution regulated by electrowetting”, *J. Heat Transf.*, 141(11), p. 111501, (2019).
- [31] Yan, R., et al., “Enhanced water capture induced with electrowetting-on-dielectric (EWOD) approach”, *Appl. Phys. Lett.*, 113(20), p. 204101, (2018).
- [32] Yan, R., et al., “Distinct condensation droplet distribution patterns under low-and high-frequency electrowetting-on-dielectric (EWOD) effect”, *Int. J. Heat Mass Transf.*, 196, p. 123269, (2022).
- [33] Bonab, M.S., Kempers, R. and Amirfazli, A., “Determining transient heat transfer coefficient for dropwise condensation in the presence of an air flow”, *Int. J. Heat Mass Transf.*, 173, p. 121278, (2021).
- [34] Ahlers, M., Buck-Emden, A. and Bart, H.J., “Is dropwise condensation feasible? A review on surface modifications for continuous dropwise condensation and a profitability analysis”, *J. Adv. Res.*, 16, pp. 1-13, (2019).
- [35] Minkowycz, W.J. and Sparrow, E.M., “Condensation heat transfer in the presence of noncondensables, interfacial resistance, superheating, variable properties, and diffusion”, *Int. J. Heat Mass Transf.*, 9(10), pp. 1125-1144, (1966).